Search task 1

- Search about One-to-One Relationship
  - Explain what a one-to-one relationship means in databases.
  - Clarify when to place the JOIN COLUMN in the first table and when to place it in the second table

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*1- **Explain what a One-to-One Relationship ?*

***1.1 Definition :-*

A one-to-one (1:1) relationship in a database means:

- Each record in Table A is associated with exactly one record in Table B, and each record in Table B is associated with exactly one record in Table A.

***1.2 Example :-*

Consider two tables:

- Employee
- Phone

Each employee has one phone, and each phone belongs to one employee.

***1.3 Implementation :-*

A one-to-one relationship is implemented using:

- Foreign Key (FK)
- UNIQUE constraint ensures that each employee is linked to only one phone

2- **Where to Place the JOIN column (Foreign Key)?

**2.1 Case 1 : Foreign Key in the Second Table :-

- Place the foreign key in the **dependent table**.

**When to use:

- **One table depends on the other**
- **One table cannot exist without the other**

بحطه في ال table الثاني لو كان ال table الثاني ده بيعتمد علي ال table الأول أو يعني ال table ده مبيبقاش ليه وجود إلا لما يكون ال table الأول ده موجود

**Example :

- **phone depends on Employee , So phone cannot exist without Employee**
- ✔️ **So Foreign key placed in Phone (dependent table)**
  - **يعني عندي هيكون فيه two tables مابينهم one-to-one relationship , وال table الأول هنا هو Employee , وال table الثاني هو phone**
  - **عشان FOREIGN KEY أحطه في أني table ? , فا هنا ال table اللي بيكون معتمد عالتاني أو ال table اللي مينفعش يكون ليه وجود إلا من غير ال table التاني فهو هنا ال Phone table**
  - **عشان ال phone فعلا بيعتمد إن يكون فيه Employee , إنو من غير Employee فال phone مش هيبقا ليه وجود**

```oracle
-- Create Employee table (parent table)
CREATE TABLE Employee (
    id NUMBER PRIMARY KEY,
    name VARCHAR2(100)
);

-- Create Phone table (dependent table)
CREATE TABLE Phone (
```

```
    id NUMBER PRIMARY KEY,
    phone_number VARCHAR2(20),

    employee_id NUMBER UNIQUE,  -- ensures one-to-one
    CONSTRAINT fk_emp FOREIGN KEY (employee_id)
        REFERENCES Employee(id)
);
```

**2.2 Case 2 : Foreign Key in the First Table :-

- **Place the foreign key in the first table when it owns or depends on the second table.**
- **Ask yourself: "Who owns who?"**
- **معناه إنك هتحط ال foreign key في table الأول فده معناه لازم يكون ال table الأول ده بيمتلك ال table الثاني , عشان تقدر تحطه في table الأول**

**When to use:

- **One table owns the other**

**Example:

- **here with this design we can put foreign key in first table when Employee "owns" the phone**
- **هنا خليت ال table الأول هنا phone وال table الثاني هو Employee فال design إختلف فهنا هسأل نفسي سؤال "مين بيمتلك التاني" فهنا ال Employee بيمتلك phone , فهنا هحط ال foreign key في phone table**

```
-- Create Phone table first
CREATE TABLE Phone (
    id NUMBER PRIMARY KEY,
    phone_number VARCHAR2(20)
);

-- Create Employee table with FK
CREATE TABLE Employee (
    id NUMBER PRIMARY KEY,
```

```
    name VARCHAR2(100),

    phone_id NUMBER UNIQUE,  -- ensures one-to-one
    CONSTRAINT fk_phone FOREIGN KEY (phone_id)
        REFERENCES Phone(id)
);
```

Another example between two cases :-

Case A: User ↔ Profile

👉 **Profile depends on User**

- **that design profile table as the second table & User table as the first table**
- **So , Foreign Key is placed in Profile table (Second table)**

👉 **User Owns the profile**

- **that design profile table as the first table & User table as the second table**
- **So , Foreign Key is placed in Profile table (First table)**

- **فهي بتختلف علي حسب design بتاع مين محطوط قبل التاني , فبسأل نفسي بأول صيغه إني table بيكون معتمد عال table تاني عشان يبقا ليه وجود , فلو هتحط ال foreign key بالصيغه دي فلازم يكون مكان ال dependant table هو second table و another table هو first table وال foreign key بحطه في second table**
- **ولو عاوزاها بصيغه إني Table بيمتلك التاني ؟ , فلو هتحط ال foreign key بالصيغه دي فلازم يكون مكان ال table اللي بيمتلك هو first table و table المملوك هو Second table وال foreign key بحطه في first table**